MCASEN: Boosting Referring Remote Sensing Image Segmentation With Multimodal Semantic Guided Decoding and Recursive Multiscale Fusion

Chongyang Li, Wenkai Zhang, *Member, IEEE*, Hanbo Bi, *Graduate Student Member, IEEE*, Jihao Li, *Member, IEEE*, Qiwei Yan, Chen Wang, Zhengyi Wang, *Member, IEEE*, Xian Sun, *Senior Member, IEEE*, and Hongqi Wang, *Member, IEEE*

***Abstract*—Referring remote sensing image segmentation (RRSIS) aims to segment the target described by textual descriptions in remote sensing (RS) images. Existing methods suffer from insufficient utilization of multimodal contextual information due to the lack of textual semantic guidance during the decoding stage, while their coarse multiscale feature fusion strategies lead to information loss and spatial structure distortion. To mitigate these limitations, we propose the multimodal context-aware semantic enhancement network (MCASEN). MCASEN comprises three key components. First, the metacognitive recursive fusion network (MCRFNet) leverages bidirectional information flow inspired by the human visual cortex to incorporate cross-hierarchical recursive feedback into multiscale feature fusion, allowing high-level semantics to iteratively refine low-level spatial details and thereby mitigating information loss. Second, the spatial perception fusion decoder (SPFD) addresses the absence of textual context during decoding by hierarchically reconstructing multiscale multimodal features and introducing adaptive rotational convolution, which preserves orientation invariance and enhances spatial structure. Finally, the multidimensional guidance fusion module (MGFM) strengthens visual feature representations through multimodal semantic enhancement and the integration of multiscale and multidirectional visual cues. We evaluated the proposed method on two public RRSIS datasets, where our method consistently achieved state-of-the-art (SOTA) performance across multiple metrics compared to existing approaches.**

***Index Terms*—Multimodal context-aware semantic enhancement network (MCASEN), referring remote sensing image segmentation (RRSIS), remote sensing (RS) multimodal learning.**

Received 19 August 2025; revised 3 November 2025 and 4 December 2025; accepted 31 December 2025. Date of current version 27 January 2026. This work was supported in part by Beijing Natural Science Foundation under Grant 4254108 and in part by the National Natural Science Foundation of China (NSFC) under Grant 62431020. *(Corresponding author: Wenkai Zhang.)*

Chongyang Li, Hanbo Bi, Qiwei Yan, and Xian Sun are with the Aerospace Information Research Institute and the Key Laboratory of Target Cognition and Application Technology (TCAT), Chinese Academy of Sciences, Beijing 100190, China, also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: lichongyang22@mails.ucas.ac.cn; bihanbo21@mails.ucas.edu.cn; yanqiwei22@mails.ucas.ac.cn; sunxian@aircas.ac.cn).

Wenkai Zhang, Jihao Li, Chen Wang, Zhengyi Wang, and Hongqi Wang are with the Aerospace Information Research Institute and the Key Laboratory of Target Cognition and Application Technology (TCAT), Chinese Academy of Sciences, Beijing 100190, China (e-mail: zhangwk@aircas.ac.cn; lijh@aircas.ac.cn; wangzhengyi@aircas.ac.cn; wiecas@sina.com).

Digital Object Identifier 10.1109/TGRS.2026.3651486

I. Introduction

REFERRING remote sensing image segmentation (RRSIS) has emerged as a new research direction in the field of remote sensing (RS) image processing in recent years [1], [2], [3]. The task aims to segment the described target in RS images based on textual descriptions. RRSIS plays a crucial role in various application scenarios, including natural disaster prevention [4], [5], agricultural monitoring [6], [7], and urban infrastructure planning [8], [9], among others [10], [11], [12].

Compared to referring segmentation tasks in natural scenes [13], [14], [15], [16], RRSIS presents greater challenges due to the inherent complexity of RS images [17], [18]. These images exhibit significant spatial scale variations and multiangular perspectives [19], [20], [21], [22], leading to object scale shifts and rotational diversity. In contrast to the relatively simple backgrounds and fixed viewpoints in natural images [23], [24], RS scenes involve heterogeneous land covers and cluttered contexts, which severely hinder the performance of traditional referring segmentation methods [25], [26], [27], [28], [29]. In recent years, many innovative RRSIS models have been developed based on the unique characteristics of RS imagery. Yuan et al. [1] introduced RRSIS and proposed the language-guided cross-scale enhancement (LGCE) module to address segmentation challenges of small and scattered objects in natural images by adaptively enhancing multiscale visual features using linguistic cues. Liu et al. [2] proposed the rotational multiscale interaction network (RMSIN) to improve segmentation results in aerial images, strengthening multiscale features through intrascale and interscale interactions and using adaptive rotational convolution to handle varying object orientations.

Although recent advances have significantly improved RRSIS, existing approaches still suffer from two key limitations. First, as illustrated in Fig. 1(a), most models employ a decoder that is isolated from textual information during mask prediction. This absence of textual context weakens semantic alignment between visual and linguistic modalities, limiting the exploitation of complementary multimodal cues and reducing segmentation accuracy, particularly for fine-grained or visually similar objects. Second, as shown

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

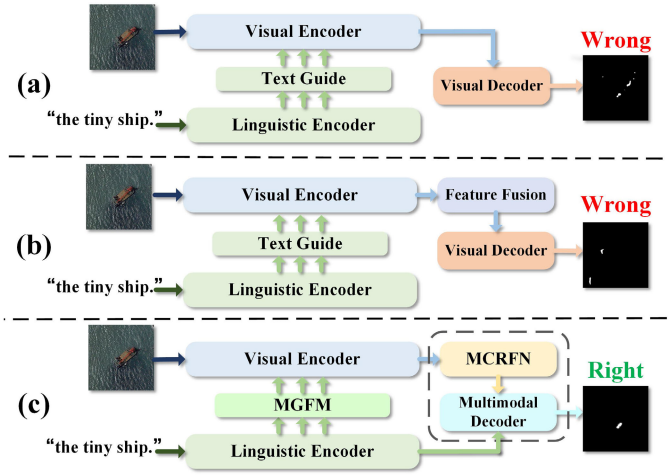


Fig. 1. Two commonly used frameworks in current RRSIS tasks, as well as our proposed MCASEN, are illustrated. In frameworks (a) and (b), the models suffer from a lack of textual contextual information during the decoding stage and handle multiscale features in a relatively coarse manner, resulting in limited segmentation performance. In contrast, our proposed MCASEN in (c) introduces MCRFNet to refine the processing of multiscale features and employs a multimodal decoder to address the problem of severe textual context information loss during decoding, thereby achieving superior segmentation performance.

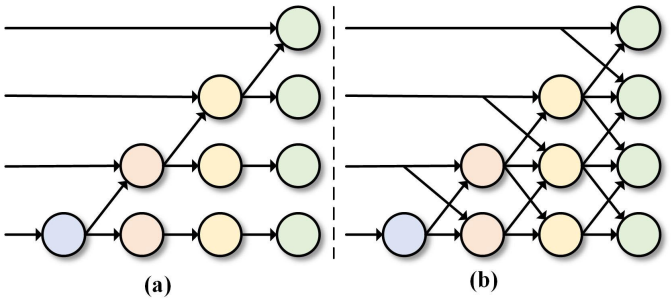


Fig. 2. Differences in information flow between the FPN and our proposed MCRFNet. In (a), the information flow within the FPN structure is limited to a bottom-up pathway, lacking feedback from higher level features, which leads to information loss. In contrast, (b) shows that MCRFNet introduces a cross-level recursive feedback mechanism that compensates for information loss during the feature transmission process.

in Fig. 1(b), some methods enhance multiscale features by incorporating FPN [30] or stacked transformers [31] between the encoder and decoder. While such designs can improve feature representation, FPN often ignores substantial differences across scales, causing information loss and redundancy, whereas transformers may disrupt the spatial integrity of multiscale features, leading to spatial information degradation. In summary, current methods are constrained by the lack of textual semantic guidance in the decoding stage, which hinders full multimodal integration, and by coarse multiscale fusion strategies, which cause information loss and spatial structure distortion.

To address the aforementioned issues, as shown in Fig. 1(c), we propose the multimodal context-aware semantic enhancement network (MCASEN). In MCASEN, we introduce the spatial perception fusion decoder (SPFD) and metacognitive recursive fusion network (MCRFNet) to tackle the problems of missing textual context information during the decoding stage and information loss caused by multiscale feature fusion, respectively.

Specifically, to refine multiscale feature fusion and enhance segmentation performance, inspired by the hierarchical feature transmission and feedback mechanism of the human visual cortex [32], we propose the MCRFNet. In the human visual system, information flows not only from lower level perceptual areas to higher level regions, but high-level semantic information also provides feedback to lower level perceptual areas. This bidirectional information flow mechanism enhances the collaborative perception of global and local information. As shown in Fig. 2, unlike the traditional FPN method, which only facilitates upward feature flow, MCRFNet introduces a cross-level recursive feedback mechanism that, while transmitting high-level features, simultaneously refines the spatial details of low-level features. This feedback mechanism significantly reduces the information loss caused by differences in feature granularity across scales during the fusion process, effectively improving segmentation performance.

To address the issue of missing textual context information during the decoding process, we propose the SPFD. Since textual features are serialized data while visual features possess spatial structure, visual features are often converted into serialized data during multimodal reasoning, leading to the loss of spatial structure information. However, spatial structure is crucial for the decoding process in RRSIS tasks, which is why most existing methods use purely visual decoders. SPFD is introduced to resolve this issue. It enhances multimodal feature quality by performing hierarchical structure restoration on multiscale multimodal features after multimodal reasoning. In addition, considering the diverse orientations of objects in RS images, SPFD utilizes the rotation-invariant properties of adaptive rotation convolution (ARC) to decode visual features into the final mask, effectively improving decoding quality.

In addition, we propose the multidimensional guidance fusion module (MGFM) to optimize the visual feature extraction process. MGFM deeply integrates visual and textual features and enhance the quality of visual features through three independent paths: first, it strengthens the multimodal semantic association through fusion with textual features; second, it captures multiscale and multiangular visual information using convolutional kernels of different sizes and shapes; and finally, it performs a weighted fusion of the three feature paths to obtain the final visual representation.

Overall, our contributions can be outlined as follows.

1) We propose MCASEN, a novel multimodal framework for RRSIS, which fully exploits multimodal contextual cues to enhance semantic alignment between visual and textual modalities.
2) To address the absence of textual guidance in the decoding stage, we design the SPFD, which reintegrates textual semantics during hierarchical reconstruction and employs adaptive rotational convolution to preserve orientation invariance and improve fine-grained segmentation.
3) To mitigate information loss and spatial distortion in multiscale fusion, we introduce the MCRFNet, which

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

incorporates cross-level recursive feedback inspired by the human visual cortex, enabling high-level semantics to refine low-level spatial details for more effective feature integration.

4) We develop the MGFM, which further enhances visual representations by integrating multimodal semantic features with multiscale and multidirectional visual information.
5) Extensive experiments on two public RRSIS datasets demonstrate that MCASEN consistently outperforms existing approaches across multiple evaluation metrics, establishing a new state of the art.

The structure of this article is organized as follows. Section II presents the related work on RRSIS. Section III elaborates in detail on our proposed MCASEN. Section IV provides comprehensive experiments and analyses. Finally, Section V wraps up this study and explores potential avenues for future research.

II. Related Works

A. Referring Image Segmentation

Referring image segmentation aims to localize target objects at the pixel level based on both an image and a query text. Unlike other multimodal tasks, this task is more complex because it requires the model to deeply understand both textual and visual data. Furthermore, it demands efficient multimodal reasoning to accurately segment the target areas within the image. Early methods in referring image segmentation relied on convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to extract visual and linguistic features, respectively. These features were then fused through simple concatenation to combine the multimodal data, ultimately producing the final segmentation mask. Li et al. [33] introduced the recursive refinement network (RRN), which takes advantage of the feature pyramid structure inherent in CNNs to capture multiscale semantic information. The network iteratively refines the segmentation mask through a recursive process, improving its accuracy with each iteration. In another approach, Liu et al. [14] developed a convolutional multimodal LSTM model that jointly processes language and image data. By encoding the sequential interactions between words, visual features, and spatial information, this model enhances segmentation performance. Nagaraja et al. [34] proposed a technique that integrates contextual information between objects, learning the probability of the referring expression through LSTM. By utilizing multiple instance learning (MIL), the model identifies contextual regions, thereby enhancing segmentation performance.

Early methods primarily focused on feature representations from a single modality, neglecting the differences between modalities and failing to fully consider the complex complementary relationship between language expressions and images. Subsequent research has been mainly dedicated to designing sophisticated alignment techniques between image and text, aiming to learn higher quality multimodal representations. Ye et al. [35] introduced a cross-modal self-attention (CMSA) module to efficiently capture long-range dependencies between language and visual features. They also designed a gated multilayer fusion module to selectively integrate CMSA features from various layers. Jing et al. [15] developed the LTS framework, which initially performs feature extraction and fusion across visual and textual inputs. The model then leverages positional cues and multimodal interactions to pinpoint the target, followed by a streamlined segmentation module that produces the final mask. Hu et al. [36] introduced the bidirectional relational inference network (BRINet), which develops a bidirectional cross-modal attention mechanism by employing both visual-guided language attention and language-guided visual attention. In addition, they created a gated bidirectional fusion module (GBFM) that facilitates the effective integration of multilevel features, ensuring precise modeling of the dependencies between vision and language. Building on the success of contrastive language-image pretraining (CLIP) [37], Wang et al. [16] developed an end-to-end framework for referring image segmentation (CRIS), driven by CLIP. By utilizing vision-language decoding alongside contrastive learning techniques, the framework achieves precise text-to-pixel alignment, enabling effective transfer of multimodal knowledge and leading to significant improvements in segmentation accuracy.

Transformers [31] have recently achieved remarkable success in referring image segmentation tasks. The language-aware visual transformer (LAVT) model [38] improves segmentation accuracy by fusing language and visual features early on at the intermediate layers of the visual transformer encoder. With the transformer encoder's strong cross-modal alignment ability, LAVT outperforms existing state-of-the-art (SOTA) approaches on several benchmark datasets. In addition, Kim et al. [39] introduced ReSTR, a convolution-free model for referring image segmentation, which employs a transformer encoder to extract both visual and language features. Through its self-attention mechanism, ReSTR seamlessly integrates the two modalities, effectively capturing long-range dependencies.

However, unlike natural images, RS images often cover a wide range of ground objects with diverse spatial scales, complex geometries, and varied orientations. These properties of RS images constrain the effectiveness of the previously mentioned methods in the RRSIS task.

B. Visual Grounding and Referring Image Segmentation in RS Images

RS visual grounding (RSVG) focuses on identifying the bounding box of a target object from an RS image and its corresponding query text. In comparison to visual grounding tasks in natural environments, RSVG tasks often involve more complex spatial relationships in the query text, and the targets in RS images tend to be less prominent, making this task more challenging. Sun et al. [40] introduced the concept of RSVG and provided the RSVG dataset. They developed GeoVG, a method that integrates a language encoder, an image encoder, and a fusion module. By employing a geospatial relationship graph and an adaptive region attention mechanism, GeoVG effectively tackles the challenges of large-scale scenes and geospatial information in RS images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

Zhan et al. [41] introduced the DIOR-RSVG dataset and proposed a transformer-based multigranularity vision-language fusion module. This module addresses large-scale variations and cluttered backgrounds in RS images by leveraging multiscale visual features and multigranularity text embeddings. Li et al. [42] presented a novel RSVG framework called language-guided progressive visual attention (LPVA). Through the use of a progressive attention (PA) module and a multilevel feature enhancement (MFE) decoder, LPVA enables precise attention adjustments for the target object, resolving issues related to insufficient interaction between visual and textual features during extraction. In addition, Li et al. [43] introduced the query-aware multimodal fusion network (QAMFN), which tackles the problem of missing early complementary information between image and text by incorporating a query-guided visual attention (QGVA) mechanism early in the visual encoder. Moreover, QAMFN includes a text semantic attention-guided mask (TAM) module to enhance the model's ability to fuse multimodal information, ultimately improving the quality of the integrated features.

RRSIS is a relatively new area of research that has gained attention in recent years. The field is still in its infancy, with limited studies available for reference. Yuan et al. [1] were the first to introduce the concept of RRSIS and developed a benchmark dataset named RefSegRS. To address the segmentation challenges of small and dispersed objects in RS images, they proposed an LGCE module, integrated within the LAVT framework [38]. The LGCE module utilizes language information as guidance to enhance segmentation accuracy for smaller targets by combining both deep and shallow features. Liu et al. [2] presented the RMSIN to tackle the issues related to spatial variation and rotational diversity in aerial imagery. RMSIN incorporates two key components, an intrascale interaction module (IIM) and a cross-scale interaction module (CIM), within the LAVT structure to enable effective feature fusion across different scales. Furthermore, to address the challenges posed by rotational variations in object orientations, the RMSIN decoder includes ARC, which improves the network's ability to capture and represent the features of rotated objects, thus enhancing the model's segmentation performance in RS environments.

III. Methods

In this section, we provide a detailed introduction to the proposed MCASEN. MCASEN consists of four key components: a linguistic encoder, a visual encoder equipped with MGFMs, an MCRFNet, and a SPFD. First, in Section III-A, we present an overview of the overall architecture of MCASEN. In Section III-B, we elaborate on the detailed structure of MCRFNet. Section III-C delves into the design of SPFD. Finally, Section III-D provides an in-depth description of the MGFM embedded within the visual encoder.

A. Overview

The RRSIS task aims to segment the target referred to by a query text within an RS image, based on both the textual query and the visual content of the image. Existing approaches suffer from two primary limitations: first, the lack of textual semantic guidance during the decoding stage leads to insufficient multimodal information fusion; second, the relatively coarse multiscale feature fusion strategies often result in information loss and the disruption of spatial structures. These deficiencies severely constrain the accuracy of current RRSIS methods. To address these issues, we propose the MCASEN. As illustrated in Fig. 3, MCASEN first utilizes a pretrained BERT [44] as the linguistic encoder to transform the query text into textual features. Meanwhile, a pretrained Swin Transformer [45] is employed as the visual encoder to extract multiscale visual features from the RS image. Furthermore, the visual encoder is equipped with the MGFM, which adjusts the visual features under the guidance of textual information to emphasize regions that are highly relevant to the query. In addition, it adaptively modifies the receptive field to enhance the network's ability to capture objects at different scales.

Subsequently, the multiscale visual features are fused within the MCRFNet. MCRFNet introduces a cross-hierarchical recursive feedback mechanism, which enables high-level features to guide the correction of spatial details in lower level features. This mechanism effectively mitigates the issues of information loss and detail degradation caused by feature discrepancies in existing methods, thus enhancing the network's ability to handle complex RS scenes. The fused multiscale features, together with the textual features, are then input into the SPFD. SPFD initially aligns and maps the visual and textual features into a unified feature space via prealignment and linear projection. Subsequently, multimodal reasoning is conducted through the multimodal context-aware fusion module (MCAFM), producing high-quality multimodal fused representations. These token-based outputs are then reconstructed into spatially structured multimodal features, which are further refined through the hierarchical structure restoration module (HSRM) to propagate multimodal information across multiscale features while preserving spatial integrity. Finally, to effectively handle the diverse orientations of objects in RS scenes, the ARC is utilized to enhance the network's capability of capturing orientation-invariant features. The resulting multiscale multimodal features are then decoded to produce the final segmentation mask.

B. MetaCognitive Recursive Fusion Network

To further refine the multiscale feature fusion process and enhance segmentation performance, we propose the MCRFNet, inspired by the hierarchical feature transmission and feedback mechanisms observed in the human visual cortex. As illustrated in the upper right corner of Fig. 3, MCRFNet introduces a cross-hierarchical recursive feedback mechanism during multiscale feature fusion, aiming to achieve effective high-level feature propagation and bottom-up correction. MCRFNet progressively integrates features from different levels through bottom-up information accumulation, while the feedback mechanism guides high-level features to refine the spatial details of lower level features, thereby significantly improving the perception of multiscale objects, complex boundaries, and fine-grained details.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

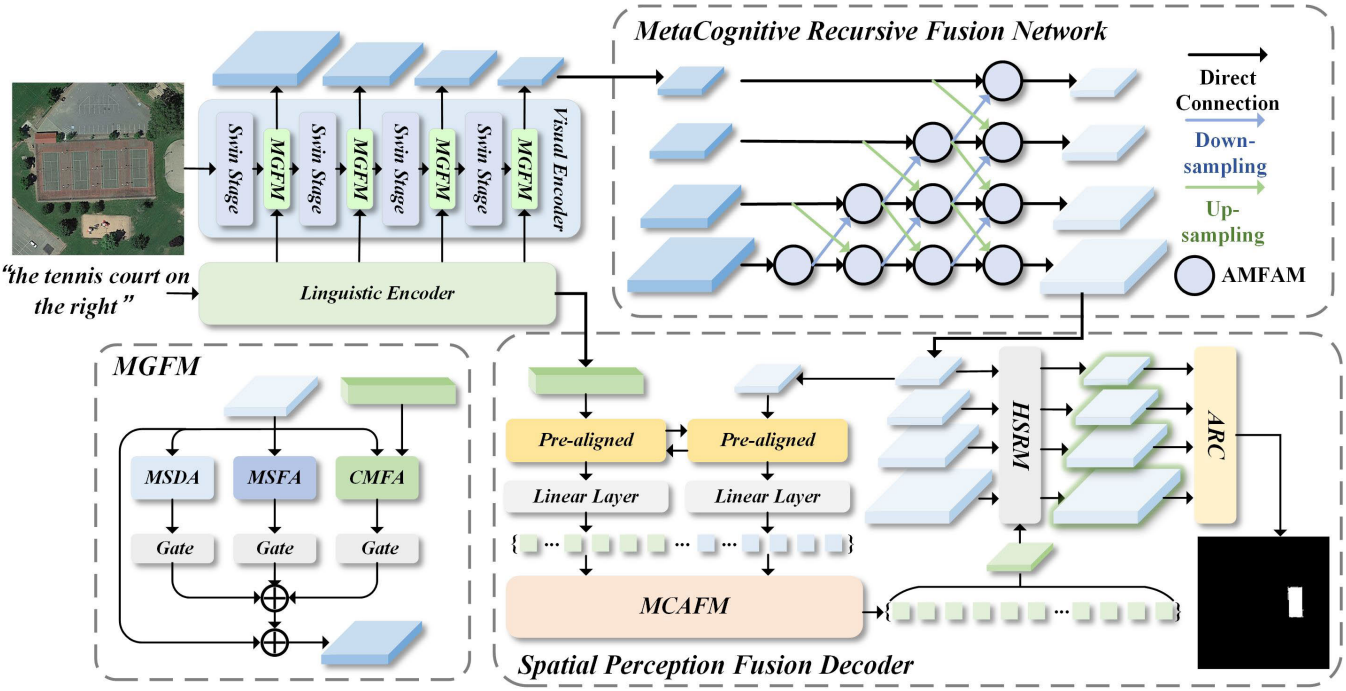


Fig. 3. Overall architecture of our proposed MCASEN. MCASEN is composed of five principal components: a linguistic encoder, a visual encoder with the MGFM, the MCRFNet, and the SPFD. The linguistic encoder transforms the query text into high-level textual representations. The visual encoder, which incorporates a pretrained Swin Transformer integrated with the MGFM, is designed to extract high-quality visual features guided by textual semantics. The MCRFNet introduces a cross-hierarchical recursive feedback mechanism to facilitate comprehensive multiscale feature interactions. Finally, the SPFD fuses textual context information with multiscale features to generate the final segmentation mask.

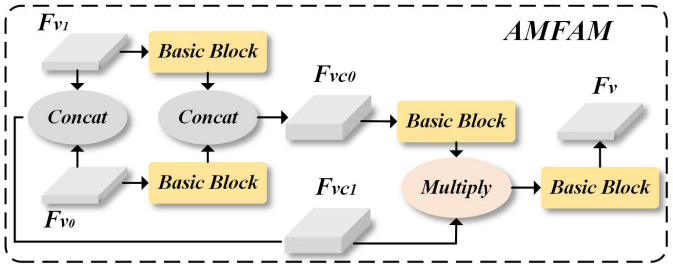


Fig. 4. Illustration of our AMFAM. In the diagram, the gray squares represent visual feature maps, all of which share the same dimensionality.

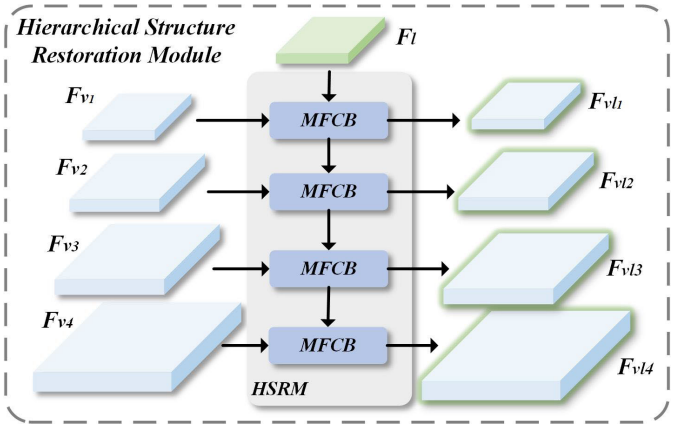


Fig. 5. Overall architecture of our proposed HSRM. HSRM progressively broadcasts the inferred multimodal information into multiscale features through MFCB at each level, enabling the integration of multimodal modulation while preserving the spatial structure of multiscale features.

Specifically, during the bottom-up propagation of lower level features, these features are first downsampled to match the resolution of higher level features. They are then fused using our proposed adaptive multiscale feature aggregation module (AMFAM), which integrates fine-grained spatial information into the high-level representations. In the top-down feedback process, the high-level features are upsampled to align with the resolution of lower level features and subsequently fused through AMFAM, continuously injecting rich semantic information into the lower level representations. This design effectively alleviates the issues of information loss and insufficient detail preservation caused by feature discrepancies in existing methods. In the following, we will provide a detailed explanation of how AMFAM performs feature fusion.

As illustrated in Fig. 4, $F_{v0} \in \mathbb{R}^{C\times H\times W}$ and $F_{v1} \in \mathbb{R}^{C\times H\times W}$ are first processed individually through a Basic Block to be mapped into a unified feature space, thereby reducing the discrepancies among features from different hierarchical levels. The specific computation formula for the basic block is as follows:

$$F_{\text{out}} = \text{RELU}\left(F_{\text{in}} + \text{BN}\left(\text{Conv}\left(\text{BN}\left(\text{Conv}\left(F_{\text{in}}\right)\right)\right)\right)\right) \quad (1)$$

where F_{out} denotes the output feature of the basic block, while F_{in} represents the input feature.

Subsequently, the mapped feature maps are concatenated along the channel dimension to obtain F_{vc0}. Meanwhile, the original inputs F_{v0} and F_{v1} are also concatenated along the channel dimension to form F_{vc1}, thereby preserving more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

original information to facilitate subsequent modulation. Next, F_{vc0} is further processed by a basic block to extract fused features and enhance semantic representation capabilities. It is then element-wise multiplied with F_{vc1} to achieve adaptive feature-level weighting and modulation. Finally, the resulting feature map after multiplication is refined by another basic block to produce the final fused output F_v. The detailed computational formulations are presented as follows:

$$F_v = \text{BasicBlock}\left(F_{vc1} \otimes \text{BasicBlock}\left(F_{vc0}\right)\right) \tag{2}$$

where $\otimes$ denotes element-wise multiplication.

C. Spatial Perception Fusion Decoder

To address the issue of missing textual contextual information during the decoding process, we propose the SPFD. The structure of SPFD is illustrated in the bottom-right corner of Fig. 3. Unlike conventional multimodal reasoning approaches that disrupt the spatial structure of visual features, SPFD adopts an "infer-then-broadcast" strategy, effectively preserving the spatial integrity of visual features while integrating multimodal information during decoding. Specifically, SPFD first performs a prealignment of visual and textual features to mitigate the substantial heterogeneity between modalities, which could otherwise interfere with subsequent reasoning. Then, through the MCAFM, textual information is deeply integrated with visual features to produce serialized fused representations. These representations are progressively introduced into the multiscale spatial structure of visual features via the HSRM, ensuring that textual information enhances the semantic relevance of visual features while better preserving their spatial details. Finally, considering the diverse rotation angles of objects in RS imagery, we employ multilayer ARC to decode the multiscale fused features into the final mask. Overall, SPFD not only addresses the issue of missing textual information during decoding but also provides a more natural and efficient solution for joint visual-textual reasoning. In the following, we provide a detailed explanation of each component of SPFD. The ARC used in our approach is the adaptive rotated convolution-based oriented-aware decoder proposed in [2]. As [2] provides a comprehensive explanation, we omit a detailed description of the decoder's structure in this article.

The input to SPFD consists of multiscale visual features F_{v1}, F_{v2}, F_{v3}, F_{v4}, and textual features F_l. We begin by performing multimodal feature fusion using the top-level visual feature F_{v1} and the textual feature F_l to obtain multimodal modulation information. Specifically, F_{v1} and F_l are first prealigned through a prealignment module, which comprises a linear layer followed by a cross-attention layer. The features are initially projected into token representations via the linear layer, after which the cross-attention layer incorporates information from the other modality to achieve alignment and fusion. The prealigned visual and textual features are then independently mapped into a unified feature space through additional linear projections. Finally, the visual feature tokens $f_v \in \mathbb{R}^{N_v \times l}$ and textual feature tokens $f_l \in \mathbb{R}^{N_l \times l}$ are concatenated to form the following sequence:

$$\boldsymbol{f} = \left[\underbrace{\boldsymbol{f}_v^1, \boldsymbol{f}_v^2, \ldots, \boldsymbol{f}_v^{N_v}}_{\text{visual tokens } f_v}, \underbrace{\boldsymbol{f}_l^1, \boldsymbol{f}_l^2, \ldots, \boldsymbol{f}_l^{N_l}}_{\text{textual tokens } f_l}\right]. \tag{3}$$

Then, the concatenated feature $f \in \mathbb{R}^{(N_v+N_l)\times l}$ is fed into the MCAFM for multimodal reasoning, aiming to fully exploit the latent correlations between visual and textual features. The MCAFM consists of six pretrained transformer encoder layers, whose parameters are initialized from the classical RSVG model MGVLF. The MCAFM produces an output f_{vl}, which is a serialized multimodal feature. We then perform an inverse operation of the visual feature serialization process to convert the serialized multimodal feature into a spatially structured multimodal feature F_l, which is subsequently utilized for feature fusion.

The HSRM is employed to effectively integrate the multimodal information from F_l into multiscale visual features, thereby enhancing the representation of text–image correlations within the visual features. Specifically, as shown in Fig. 5, F_{v1} is first fused with F_l via the multimodal feature cross-dim broadcaster (MFCB) module to obtain F_{vl1}, enabling the initial integration of visual and textual information. Subsequently, F_{vl1} undergoes feature enhancement and upsampling to generate F_{vlu1}, which is then fused with F_{v2} to further refine the multimodal feature representation. After fusion through the MFCB module, the output F_{vl2} undergoes feature enhancement and upsampling again, ensuring that multimodal information is preserved and optimized at each stage. Through this iterative process, multiscale multimodal features F_{vl1},F_{vl2},F_{vl3}, and F_{vl4} are obtained, effectively combining multimodal and multiscale spatial information, and making them suitable for subsequent decoding tasks.

The MFCB module plays a crucial role in this process, performing multimodal feature fusion, feature enhancement, and upsampling to ensure the comprehensiveness and richness of the feature representation. The MFCB module consists of two key components: the Fusion Block and the Upsample Block. The primary goal of the MFCB is to inject multimodal information into multiscale features, thereby enhancing the expression of joint text-image features. In the ith feature fusion step, the input features F_{vi} and$F_{vlu(i-1)}$ are first concatenated along the channel dimension. The concatenated feature is then processed by the Fusion Block, which applies convolution, batch normalization, and ReLU activation to refine the features. The computation of the fusion block is defined as

$$F_{vli} = \text{ReLU}\left(\text{BN}\left(\text{Conv}\left(\text{Concat}\left(F_{vi}, F_{vlu(i-1)}\right)\right)\right)\right). \tag{4}$$

The convolution operation, followed by batch normalization and ReLU, refines the feature map, adding nonlinearity to enhance its representational power. Next, the output F_{vli} from the fusion block is processed by the upsample block, which increases the feature map resolution. This step is essential for aligning multiscale features, ensuring they can be effectively

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

fused at higher layers in the network. The computation of the upsample block is expressed as

$$F_{vlui} = \text{Upsample}\left(\text{FusionBlock}\left(\text{CS}\left(\text{Conv}\left(F_{vli}\right)\right)\right)\right). \quad (5)$$

The Conv layer generates feature maps that are suitable for grouping. The channel shuffle (CS) operation is then applied to rearrange the channels, which improves the information exchange between different groups of channels. This rearrangement increases the diversity of the feature map's channel interactions, allowing for more complex and enriched feature representations. After this operation, the FusionBlock refines the features further, and the upsample operation enlarges the feature map resolution, aligning it with the requirements of the next layer.

D. Multidimensional Guidance Fusion Module

To enhance the quality of visual features extracted by the visual encoder, we propose a novel MGFM. MGFM strengthens visual representations through three independent processing branches. First, the cross-modal feature alignment (CMFA) mechanism leverages textual features to guide visual feature learning, thereby enhancing the semantic correlation between modalities. Second, the multiscale feature aggregation (MSFA) module is designed to enrich the expression of visual information across multiple spatial scales. Third, recognizing the diverse orientations of objects in RS images, we introduce a multishape directional aggregation (MSDA) module to capture directional visual cues from various angles. The enhanced features from all three branches are subsequently projected into a unified feature space through individual gate modules, normalized to an appropriate value range, and then fused via summation. Finally, the fused visual features are integrated with the original visual inputs through a residual connection to obtain the final output.

Next, I will present a comprehensive overview of the CMFA module, MSFA module, MSDA module, and gate module.

1) CMFA Module: The CMFA module is designed to leverage textual features to guide visual features, enhancing the representation of the parts in visual features that are relevant to textual features. Given the visual feature F_{vj} and the textual feature F_l, the module first employs F_{vj} as the query, F_l as the key and value, and applies scaled dot-product attention to obtain the fused multimodal feature A_j

$$A_j = \text{attention}\left(F_{vj}W_q, F_lW_k, F_lW_v\right) \quad (6)$$

where W_q, W_k, and W_v are the parameter matrices of the linear layers. Then, the multimodal feature A_j is applied to the initial visual feature F_{vj} to obtain the text-guided visual feature F_{vl}

$$F_{vl} = W_a\left(W_bA_j \otimes W_cF_{vj}\right) \quad (7)$$

where W_a, W_b, and W_c are the mapping matrices.

2) MSFA Module: In RS images, object scales are inherently imbalanced, and there exists a significant disparity between local and global information. To effectively capture objects of varying scales while enhancing the influence of global context on local feature representation, we propose the MSFA module. This module expands the receptive field through multiscale convolutions, adaptively adjusts feature weights via an attention mechanism, and integrates both local and global information. As a result, it enhances feature extraction capabilities in RS imagery and improves object differentiation.

Assuming that the visual feature input to the MSFA module is denoted as F_i, we first compute the mean representation of F_i, along the channel dimension, obtaining F_{mean}. Next, F_{mean} is processed using convolutional kernels of different sizes to capture visual features with varying receptive fields. To ensure that these visual features maintain the same spatial dimensions, appropriate padding values are applied. Finally, these multiscale visual features are summed to generate the visual feature mask F_{mask}, which integrates information from different receptive fields. The specific computation formula is as follows:

$$F_{\text{mask}} = \text{RELU}\left(\text{BN}\left(\text{Conv}_{5\times5}\left(F_{\text{mean}}\right)\right) + \text{BN}\left(\text{Conv}_{3\times3}\left(F_{\text{mean}}\right)\right) + \text{BN}\left(\text{Conv}_{1\times1}\left(F_{\text{mean}}\right)\right)\right) \quad (8)$$

where RELU represents the ReLU activation function and BN stands for batch normalization. Finally, after broadcasting F_{mask} along the channel dimension, it is element-wise multiplied with F_i to obtain the final output $\overline{F_i}$, as expressed in the following equation:

$$\overline{F_i} = F_i \otimes F_{\text{mask}} \quad (9)$$

where $\otimes$ denotes element-wise multiplication.

3) MSDA Module: To extract direction-sensitive features from RS images while simultaneously expanding the receptive field to capture more distant contextual information, we propose the MSDA module. Compared to the MSFA module, which utilizes convolutional layers of different kernel sizes, the MSDA module introduces convolutional kernels of various shapes to extract features in different orientations within RS images. In addition, it employs dilated convolutions to enhance the model's ability to perceive global information.

For the input visual feature F_{li} of the MSDA module, we first apply convolutions with different kernel sizes combined with dilated convolutions to extract features in various directions from F_{li}. The specific computation is formulated as follows:

$$X_i = \text{Conv}_{3\times3}\left(\text{Conv}_{1\times1}\left(F_{li}\right)\right) \quad (10)$$

$$X_j = \text{DilatedConv}\left(\text{Conv}_{3\times1}\left(\text{Conv}_{1\times3}\left(F_{li}\right)\right)\right) \quad (11)$$

$$X_k = \text{DilatedConv}\left(\text{Conv}_{1\times3}\left(\text{Conv}_{3\times1}\left(F_{li}\right)\right)\right) \quad (12)$$

where DilatedConv represents the dilated convolution, which enhances the receptive field by inserting gaps between kernel elements, allowing efficient multiscale feature extraction while preserving resolution. Subsequently, X_i, X_j, and X_k are concatenated along the channel dimension to form the joint feature representation X_{all}. Finally, X_{all} is fused with the initial input visual feature F_{li} to obtain the final output $\overline{F_{li}}$. The specific computation is formulated as follows:

$$\overline{F_{li}} = \text{RELU}\left(\text{Conv}_{1\times1}\left(X_{\text{all}}\right) + \text{Conv}_{1\times1}\left(F_{li}\right)\right). \quad (13)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

4) Gate Module: The gate module is designed to map features with different characteristics into a unified feature space, facilitating subsequent feature fusion. The specific calculation formula is as follows:

$$F_{\text{out}} = \text{RELU}\left(\text{Linear}\left(\text{RELU}\left(\text{Linear}\left(F_{\text{in}}\right)\right)\right)\right) \tag{14}$$

where F_{in} denotes the input features, F_{out} represents the output features, and Linear refers to the linear layer.

IV. Experiments

This section presents a thorough set of experiments to highlight the strengths of the proposed MCASEN. Sections IV-A and IV-B offer an in-depth description of the experimental configuration, datasets, and evaluation metrics used. Section IV-C presents the quantitative results of MCASEN, comparing its performance against various SOTA methods. In Section IV-D, a series of ablation studies is conducted to evaluate the contribution of each module within MCASEN. Lastly, Section IV-E showcases qualitative results, providing a visual representation of MCASEN's impressive capabilities.

A. Datasets and Evaluation Metric

1) Datasets: For training and evaluating our model, we utilized two well-established datasets in the RRSIS domain: RefSegRS [1] and RRSIS-D [2]. The RefSegRS dataset contains a total of 4420 image-text-label triplets, which encompass 285 distinct scenes and 14 different target categories. Each image in this dataset is standardized to a resolution of 512×512 pixels, with a spatial resolution of 0.13 m, ensuring fine-grained details for accurate segmentation. This dataset is widely used for evaluating segmentation models and provides a solid foundation for assessing RRSIS tasks.

In comparison, the RRSIS-D dataset is larger in scale and more diverse. It consists of 17 402 images, spanning 20 target categories and seven attributes, making it a more complex and comprehensive dataset for RRSIS research. Each image in this dataset is meticulously annotated with both a corresponding mask and referring expressions, which describe specific regions of interest. The dataset is divided into a training set with 12 181 image-text-label triplets, a validation set consisting of 1740 triplets, and a test set that includes 3481 triplets. This structure enables a thorough evaluation of model performance across multiple subsets, allowing for more robust training and testing.

2) Evaluation Metric: Building on prior work [1], [2], we utilize several evaluation metrics, including overall intersection over union (oIoU), mean intersection over union (mIoU), and precision at varying thresholds from 0.5 to 0.9 (Pr@X). oIoU measures the proportion of the intersection relative to the union over the whole test set, with a higher weight given to larger objects. On the other hand, mIoU calculates the mean IoU between the predicted results and the ground truth for each sample, ensuring that objects of varying sizes are treated uniformly. The mathematical definitions for oIoU and mIoU are as follows:

$$\text{mIoU} = \frac{1}{N}\sum_{n} I_n/U_n \tag{15}$$

$$\text{oIoU} = \left(\sum_{n} I_n\right) \Big/ \left(\sum_{n} U_n\right) \tag{16}$$

where n denotes the index of a specific sample, while N refers to the total number of samples in the dataset. I_n and U_n denote the areas of overlap and total coverage, respectively, between the predicted regions and the corresponding ground truth regions.

B. Implementation Details

We utilize the RefSegRS and RRSIS-D datasets to train and evaluate our model. During the training process, all images are resized to a uniform resolution of 480×480. The model is optimized using the AdamW [55] optimizer with a weight decay of 0.01 and an initial learning rate of 0.00005. This learning rate is adjusted using a polynomial decay schedule throughout the training. We set the batch size to 32 and train each model for 40 epochs using 8 NVIDIA GeForce RTX 3090 GPUs. For parameters that are not pretrained, Xavier initialization [56] is applied for random initialization.

Our method uses a pretrained Swin Transformer as the visual backbone, which has been trained on the ImageNet-22K dataset [57]. The multimodal transformer module is built upon the six-layer transformer encoder architecture of DETR [58], with each layer comprising eight attention heads and a 512-D hidden state, and is initialized with DETR encoder weights pretrained on the COCO 2017 dataset. For language encoding, we adopt the BERT model [44] with a hidden dimension of 768. In line with the approach in [2], we train our model using a combination of cross-entropy loss and dice loss, with the weight of the dice loss set to 0.1.

C. Comparisons With SOTA Methods

We performed comprehensive experiments on the RefSegRS and RRSIS-D datasets to evaluate the performance of the proposed MCASEN. The results show that our approach outperforms previous methods on various metrics across both datasets, achieving substantial improvements. These outcomes clearly highlight the effectiveness and superiority of our method in the RRSIS task. We selected both classical approaches and the most advanced SOTA techniques for comparison. The visual encoders used in these methods include ResNet-101 [59], and Swin Transformer [45], while the text encoders encompass LSTM [60], CLIP [37], and BERT [44]. The experimental results on the RefSegRS and RRSIS-D datasets are presented in Tables I and II, respectively. In these tables, bold numbers indicate the best performance achieved by a model for a given metric, while underlined numbers denote the second-best performance.

From the experimental results, it is evident that methods such as RRN [33], CSMA [35], LSCM [46], CMPC [47], and BRINet [36] are constrained by their limited ability to fuse multimodal features. In addition, LSTM [60] exhibits weak capability in extracting textual features, making it challenging for these methods to achieve satisfactory results in the RRSIS task. The CMPC+ [48] model simulates the human cognitive process by decomposing the RRSIS task into

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

TABLE I

EXPERIMENTAL RESULTS ON THE REFSEGRS DATASET. THE BEST PERFORMANCE WITH BOLD, AND THE SECOND PERFORMANCE IS WITH UNDERLINE

Methods	Venue	Visual Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
RRN [33]	*CVPR'18*	ResNet-101	LSTM	30.84	22.97	14.71	6.95	0.95	65.82	42.77
CSMA [35]	*CVPR'19*	ResNet-101	None	28.27	20.69	13.12	5.58	0.91	64.12	41.94
LSCM [46]	*ECCV'20*	ResNet-101	LSTM	32.38	31.26	10.21	6.17	1.25	63.08	39.14
CMPC [47]	*CVPR'20*	ResNet-101	LSTM	27.15	16.79	11.37	8.86	0.91	60.94	33.14
BRINet [36]	*CVPR'20*	ResNet-101	LSTM	22.45	15.56	10.47	4.14	1.31	60.23	32.65
CMPC+ [48]	*TPAMI'21*	ResNet-101	LSTM	52.67	48.95	30.78	15.14	5.26	69.35	55.77
ETRIS [49]	*ICCV'23*	ResNet-101	CLIP	65.25	55.06	30.10	14.22	5.30	71.94	52.38
CRIS [16]	*CVPR'22*	ResNet-101	CLIP	64.76	53.98	29.62	14.74	4.25	72.14	53.82
LAVT [38]	*CVPR'22*	Swin-B	BERT	71.44	57.40	32.14	15.41	4.51	76.46	57.74
CrossVLT [50]	*TMM'23*	Swin-B	BERT	71.16	58.28	34.51	16.35	5.06	77.44	58.84
RIS-DMMI [51]	*ICCV'23*	Swin-B	BERT	76.63	66.50	47.51	17.30	2.40	76.53	60.55
robust-ref-seg [52]	*TIP'24*	Swin-B	BERT	74.85	60.21	34.01	15.58	4.24	76.92	59.35
LGCE [1]	*TGRS'24*	Swin-B	BERT	73.58	59.77	36.43	16.29	4.68	75.15	59.52
RMSIN [2]	*CVPR'24*	Swin-B	BERT	68.74	52.17	28.07	10.46	1.76	71.59	56.03
FIANet [53]	*TGRS'24*	Swin-B	BERT	78.26	67.75	48.86	21.86	6.31	76.95	63.47
CADFormer [54]	*JSTARS'25*	Swin-B	BERT	76.46	66.73	46.38	22.35	5.89	75.62	62.85
MCASEN(Ours)	-	Swin-B	BERT	**80.48**	**71.62**	**49.22**	**23.74**	**7.01**	**78.16**	**65.43**

TABLE II

EXPERIMENTAL RESULTS ON THE RRSIS-D DATASET. THE BEST PERFORMANCE WITH BOLD, AND THE SECOND PERFORMANCE IS WITH UNDERLINE

Methods	Venue	Visual Encoder	Language Encoder	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
RRN [33]	*CVPR'18*	ResNet-101	LSTM	53.27	45.31	35.84	22.53	6.55	66.71	46.32
CSMA [35]	*CVPR'19*	ResNet-101	None	56.14	48.88	38.32	27.51	8.57	69.71	48.25
LSCM [46]	*ECCV'20*	ResNet-101	LSTM	57.82	49.12	38.11	26.94	8.02	69.56	50.94
CMPC [47]	*CVPR'20*	ResNet-101	LSTM	58.35	49.22	38.89	26.01	9.36	71.25	51.46
BRINet [36]	*CVPR'20*	ResNet-101	LSTM	59.43	50.17	39.56	28.21	9.39	70.36	50.51
CMPC+ [48]	*TPAMI'21*	ResNet-101	LSTM	58.35	49.51	38.57	25.61	8.56	70.15	51.21
ETRIS [49]	*ICCV'23*	ResNet-101	CLIP	54.84	46.77	38.06	28.15	11.52	70.46	49.69
CRIS [16]	*CVPR'22*	ResNet-101	CLIP	61.07	50.99	40.94	29.30	11.43	71.06	54.21
LAVT [38]	*CVPR'22*	Swin-B	BERT	69.52	63.63	53.29	41.60	24.94	77.19	61.04
CrossVLT [50]	*TMM'23*	Swin-B	BERT	70.38	63.83	52.86	42.11	25.02	76.32	61.30
RIS-DMMI [51]	*ICCV'23*	Swin-B	BERT	72.63	65.97	53.99	38.88	21.67	74.84	62.06
robust-ref-seg [52]	*TIP'24*	Swin-B	BERT	68.54	61.96	52.40	40.85	24.27	76.99	60.06
LGCE [1]	*TGRS'24*	Swin-B	BERT	69.32	62.83	52.71	40.68	24.36	77.14	60.38
RMSIN [2]	*CVPR'24*	Swin-B	BERT	73.97	67.16	55.50	41.71	23.93	77.11	63.44
FIANet [53]	*TGRS'24*	Swin-B	BERT	74.46	66.96	56.31	42.83	24.13	76.91	64.01
CADFormer [54]	*JSTARS'25*	Swin-B	BERT	74.20	67.62	55.59	42.37	23.59	77.26	63.77
MCASEN(Ours)	-	Swin-B	BERT	**77.39**	**70.40**	**59.02**	**44.20**	**26.53**	**79.03**	**66.46**

two stages: first, it coarsely locates candidate entities, and then it further differentiates the target entity. This approach leads to performance improvements over previous methods. ETRIS [49] incorporates CLIP [37] for text feature extraction and employs parameter-efficient fine-tuning to reduce computational resource consumption. However, this fine-tuning approach also restricts the model's expressive capacity, lowering its upper bound performance and limiting its effectiveness in complex RRSIS tasks. CRIS [16] proposes an end-to-end referring image segmentation framework based on CLIP [37]. However, due to the lack of substantial RS data in the pretrained CLIP [37] model, its performance in the RRSIS task is compromised. LAVT [38], CrossVLT [50], RIS-DMMI [51], and Robust-Ref-Seg [52] are all end-to-end referring image segmentation methods that demonstrate promising performance in the RRSIS task. Nevertheless, they fail to achieve optimal results across all metrics due to the absence of design considerations specific to RS images.

LGCE [1], RMSIN [2], FIANet [53], and CADFormer [54] are models proposed for the RRISIS task. LGCE leverages linguistic features to adaptively enhance multiscale visual representations and integrates both deep and shallow visual information. However, its handling of multiscale features is relatively coarse, which results in the loss or redundancy of scale-specific information, ultimately limiting its ability to achieve more accurate segmentation. RMSIN introduces intrascale and CIMs to integrate fine-grained visual details and employs adaptive rotational convolutions to address objects with varying orientations. Nevertheless, RMSIN performs only basic multimodal feature interactions within the visual backbone, which restricts its capacity to fully exploit multimodal contextual information for precise segmentation. FIANet proposes a fine-grained vision-language alignment module to jointly mine object and spatial information from both image and text, and incorporates a text-aware multiscale enhancement module to adaptively fuse visual features across

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EXPERIMENTAL RESULTS ON MODEL PARAMETERS AND FLOPS

Methods	RMSIN	FIANet	CADFormer	LGCE	MCASEN
Params(M)	208.93	226.49	154.85	164.28	214.82
Flops(G)	154.30	329.7	233.25	189.21	159.37

TABLE IV
ABLATION EXPERIMENT OF THE MAIN MODULES

MGFM	MCRFNet	SPFD	Pr@0.5	Pr@0.7	Pr@0.9	oIou	mIou
-	✓	✓	74.76	55.21	24.57	76.97	64.13
✓	✓	-	73.27	53.38	22.97	77.15	62.60
✓	-	✓	74.14	55.71	24.32	77.53	64.20
-	-	✓	67.79	45.52	16.96	67.05	53.95
✓	-	-	68.25	47.61	17.74	68.83	54.74
✓	✓	✓	**77.39**	**59.02**	**26.53**	**79.03**	**66.46**

different scales. However, FIANet lacks the integration of textual contextual information during the decoding stage, leading to insufficient multimodal correlation and limiting its segmentation performance. CADFormer achieves fine-grained cross-modal feature fusion through semantic mutual-guidance alignment and text-enhanced decoding; however, it overlooks the fine-grained handling of multiscale information, which limits its segmentation performance.

In the proposed MCASEN framework, we refine the multiscale feature fusion process by introducing a cross-hierarchical recursive feedback mechanism, which effectively reduces information loss and enhances segmentation performance. In addition, we incorporate a novel multimodal decoder that optimizes the reasoning process across modalities, thereby preventing the structural information in the visual feature space from being lost during multimodal inference. This design also addresses the issue of missing textual contextual information during decoding. Experimental results validate the effectiveness of our approach. As shown in Table I, MCASEN achieves a 2.2% improvement over the second-best method on the RefSegRS dataset in terms of Pr@0.5, and an 11.74% improvement compared to RMSIN, demonstrating superior segmentation accuracy. This indicates that MCASEN can more precisely locate the target referred to by the query in RS images. Furthermore, MCASEN also achieves the best performance in both oIoU and mIoU metrics, surpassing the second-best method by 0.72% and 1.96%, respectively, which confirms the higher accuracy and clearer boundaries of the segmentation masks produced by MCASEN.

Similarly, the results on the RRSIS-D dataset, as shown in Table II, further demonstrate the superiority of our method. From Pr@0.5 to Pr@0.9, MCASEN outperforms the second-best method by 2.93%, 2.78%, 2.71%, 1.37%, and 2.40%, respectively, indicating consistent SOTA performance across different precision thresholds. In terms of oIoU and mIoU, MCASEN also achieves improvements of 1.77% and 2.45%, respectively, further highlighting the model's ability to produce more accurate and finely delineated segmentation results.

In summary, the quantitative results on both datasets clearly demonstrate that the proposed MCASEN significantly advances the performance upper bounds in the RRSIS task, enabling more accurate segmentation of the target regions referred to by the input queries.

To verify that the proposed method does not introduce excessive parameters or computational overhead, we compared the number of parameters and FLOPs of MCASEN with several representative baseline models, as shown in Table III. In terms of parameter count, MCASEN is slightly higher than RMSIN, CADFormer, and LGCE, but remains lower than FIANet, which achieves the second-best performance across multiple metrics. Regarding FLOPs, MCASEN demonstrates high computational efficiency, being only marginally higher than RMSIN while lower than all other methods. These results indicate that our method achieves optimal performance without relying on overly complex computational modules, demonstrating that well-designed architectural strategies can effectively balance efficiency and accuracy.

D. Ablation Study

To assess the contribution of each module in the proposed MCASEN, we conducted a series of ablation studies using the RRSIS-D dataset. We specifically selected the test set of the RRSIS-D dataset for evaluation and measured the performance of different ablation models across five key metrics: Pr@0.5, Pr@0.7, Pr@0.9, oIoU, and mIoU.

1) Ablation Experiments for Entire Network: Table IV presents the results of the ablation experiments for the key modules in MCASEN. A checkmark indicates the presence of a specific module in the model, while its absence is denoted otherwise. When the MGFM module is excluded, the visual feature extraction process lacks guidance from textual information and fails to effectively capture multiscale and multidirectional visual cues, leading to a decline in segmentation accuracy. The absence of MCRFNet results in suboptimal utilization of multiscale information, causing a loss of fine-grained details in the multiscale features and ultimately impairing segmentation performance. Without the SPFD module, the decoding phase lacks the integration of textual contextual information, weakening the association between multimodal features and leading to a noticeable drop across evaluation metrics.

Additional experiments were also conducted under configurations where two of the three modules were removed. When the model relies solely on either MGFM or SPFD to perform multimodal fusion but lacks an effective mechanism for integrating multiscale information, its performance drops significantly across all evaluation metrics. In summary, the three proposed key modules, MGFM, MCRFNet, and SPFD, are all crucial for enhancing model performance. They collectively ensure the effective fusion of multiscale and multimodal information, enabling the model to achieve optimal segmentation results.

2) Ablation Study on the MGFM: Table V presents the results of the ablation study conducted on the submodules within MGFM, which consists of three components: MSFA, CMFA, and MSDA. The experimental results indicate that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

TABLE V
ABLATION EXPERIMENT OF THE MGFM

MSFA	CMFA	MSDA	Pr@0.5	Pr@0.7	Pr@0.9	oIou	mIou
-	-	-	74.76	55.21	24.57	76.97	64.13
-	✓	✓	76.75	57.67	26.17	77.78	65.23
✓	-	✓	75.12	56.08	25.21	77.17	65.25
✓	✓	-	76.34	56.63	24.76	76.79	64.82
✓	✓	✓	**77.39**	**59.02**	**26.53**	**79.03**	**66.46**

TABLE VI
ABLATION EXPERIMENT OF THE MGFM AT DIFFERENT STAGES

Stage1	Stage2	Stage3	Stage4	Pr@0.5	Pr@0.7	Pr@0.9	oIou	mIou (%)
-	-	-	-	74.76	55.21	24.57	76.97	64.13
-	-	-	✓	74.93	54.96	25.06	76.79	64.51
-	-	✓	✓	75.21	56.94	25.48	77.02	65.34
-	✓	✓	✓	76.64	58.06	26.21	77.89	66.41
✓	✓	✓	✓	**77.39**	**59.02**	**26.53**	**79.03**	**66.46**

TABLE VII
ABLATION EXPERIMENT OF THE MCRFNET

S1	S2	S3	Pr@0.5	Pr@0.7	Pr@0.9	oIou	mIou	PSI(ms)
-	-	-	74.14	55.71	24.32	77.53	64.20	**85.21**
-	-	✓	75.04	55.96	24.51	77.61	64.76	87.82
-	✓	✓	76.16	57.84	24.97	77.82	65.42	91.98
✓	✓	✓	**77.39**	**59.02**	**26.53**	**79.03**	**66.46**	95.47

TABLE VIII
ABLATION EXPERIMENT ON THE MULTISCALE FEATURE FUSION MODULE

Methods	Pr@0.5	Pr@0.7	Pr@0.9	oIou	mIou
FPN[30]	73.92	56.30	24.94	75.89	63.32
ASFF[61]	74.63	56.97	25.32	76.58	64.86
BiFPN[62]	75.42	57.68	25.72	77.36	65.38
CIM[2]	75.36	57.41	25.12	78.13	65.07
MCRFNet	**77.39**	**59.02**	**26.53**	**79.03**	**66.46**

removal of any individual submodule leads to a noticeable degradation in model performance, highlighting the importance of each component. When the MSFA submodule is removed, the model's ability to perceive objects with significant scale variation in RS images is weakened. As a result, it fails to effectively capture targets of different sizes, leading to reduced segmentation accuracy. In the absence of the CMFA submodule, textual information can no longer guide the process of visual feature extraction. This misalignment causes a discrepancy between the visual features extracted by the backbone and those required for subsequent multimodal reasoning, resulting in a considerable drop across all evaluation metrics. Without the MSDA submodule, the model's capacity to perceive visual information from multiple directions is compromised, which negatively impacts its segmentation capability. The complete MGFM module achieves the best performance across all metrics, clearly demonstrating the necessity of each submodule. All three components play a crucial role in enhancing the model's overall effectiveness and contribute significantly to its ability to produce accurate segmentation results.

Furthermore, to further verify the effectiveness of the MGFM module, we conducted experiments by placing MGFM after different stages of the visual backbone, as shown in Table VI. The results show that removing the MGFM after any stage leads to a decline in model performance, demonstrating the importance of deploying the MGFM at each stage. When all MGFMs were removed, both feature enhancement and textual guidance during visual feature extraction were lost, resulting in a significant drop in performance. When MGFM was placed only after the last stage, the textual guidance for visual features was insufficient, which hindered subsequent multimodal alignment and limited the model's overall capability. As the number of MGFM modules added after each stage increased, the visual features extracted by the backbone received stronger textual guidance, enabling more comprehensive capture of multiscale and multidirectional feature information, and consequently leading to continuous improvements in model performance.

3) Ablation Study on the MCRFNet: Table VII presents the results of the ablation study on MCRFNet, where S1, S2, and S3 denote the stages at which recursive feedback is applied. As illustrated in Fig. 6, configuration (a) applies recursive feedback only at the final stage (S3), configuration (b) applies it at stages S2 and S3, and configuration (c) incorporates recursive feedback at all three stages from S1 to S3. The experimental results show that when no recursive feedback is applied, MCRFNet degenerates into a standard FPN, and the model achieves only limited performance across all evaluation metrics. Introducing recursive feedback at the final stage allows high-level features to guide the fusion process, resulting in a modest improvement in performance. When recursive feedback is applied at two stages, the influence of high-level feature refinement becomes more pronounced, leading to further gains. The best results are achieved when recursive feedback is applied at every stage. In this setting, high-level features progressively refine lower level features across stages, enabling the model to produce optimal segmentation outcomes. These findings clearly demonstrate the effectiveness of the proposed recursive feedback mechanism. By introducing feedback from high-level features to progressively optimize features at different levels, the quality of multiscale feature representations is significantly enhanced, which in turn leads to improved segmentation performance.

In addition, we also compared the per-sample inference time (PSI) of the model under four scenarios. Although the recursive feedback mechanism increases the computational load, the computation in AMFAM is not complex, thus exerting little impact on the overall inference time of the model. As indicated in Table VII, each additional stage of the recursive feedback mechanism only increases the PSI by 2–3 ms.

To further validate the superiority of the proposed MCRFNet in multiscale feature fusion, we compare MCRFNet with several classical multiscale fusion methods within the multiscale feature fusion module. The experimental results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

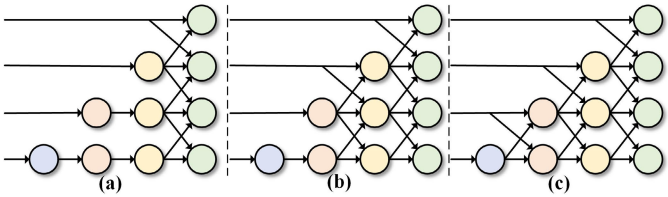


Fig. 6. Ablation settings for MCRFNet. (a) Recursive feedback is applied only at the final stage. (b) Recursive feedback is applied at the last two stages. (c) Recursive feedback is applied at every stage.

TABLE IX
ABLATION EXPERIMENT OF THE SPFD

Pre-alignment	MCAFM	HSRM	Pr@0.5	Pr@0.7	Pr@0.9	oIou	mIou
-	-	-	73.27	53.38	22.97	77.15	62.60
-	✓	✓	76.54	57.48	25.57	78.14	65.43
✓	-	✓	74.68	55.12	23.86	77.35	63.18
✓	✓	-	75.64	55.68	24.46	77.76	63.49
✓	✓	✓	**77.39**	**59.02**	**26.53**	**79.03**	**66.46**

are presented in Table VIII. As shown, FPN and its variants ASFF and BiFPN ignore the bottom-up feedback process during multiscale feature fusion, resulting in the loss of low-level detailed information and consequently suboptimal performance. Although BiFPN introduces a certain degree of feedback, it only achieves the second-best results across several metrics. CIM, a transformer-based multiscale fusion method, suffers from the loss of spatial information in the visual feature space during the fusion process, which limits its effectiveness. In contrast, the proposed MCRFNet not only preserves the top-down guidance of high-level semantic features but also incorporates the bottom-up feedback of low-level details. As a result, it achieves the best performance across all metrics, demonstrating the effectiveness of the proposed approach.

4) Ablation Study on the SPFD: Table IX presents the results of the ablation study conducted on SPFD, which consists of three primary components. First, a prealignment process is applied to reduce the heterogeneity between multimodal features. Next, MCAFM is used to perform reasoning on the preliminarily fused multimodal features. Finally, HSRM broadcasts the integrated multimodal information to multiscale features. The experimental results indicate that omitting the prealignment step prevents the model from reducing the heterogeneity between visual and textual features before multimodal reasoning. This negatively affects the reasoning process and limits segmentation performance. When MCAFM is removed, the SPFD module lacks the ability to perform multimodal reasoning. In this case, the prealignment alone is insufficient for generating high-quality multimodal features, resulting in weaker decoding performance. When HSRM is excluded, the SPFD module only integrates textual information into high-level features without broadcasting it to features at multiple scales. As a result, the representation of multimodal information across different scales is restricted, which limits the effectiveness of the decoder. The complete SPFD module achieves the best performance across all evaluation metrics, clearly demonstrating that each of its components is indispensable for optimal model performance.

TABLE X
ABLATION EXPERIMENT OF THE NUMBER OF TRANSFORMER ENCODERS IN MCAFM

Encoder Number	Pr@0.5	Pr@0.7	Pr@0.9	oIou	mIou
2	74.56	54.83	23.47	75.86	61.94
4	75.82	57.04	25.48	78.01	62.57
6	77.39	59.02	26.53	79.03	66.46
8	78.14	58.38	25.97	79.11	65.85
10	78.09	59.43	25.85	79.48	66.70

As shown in Table X, we conducted an ablation study to investigate the impact of the number of transformer encoders in the MCAFM. The results indicate that when the number of transformer encoders is set to 2 or 4, the multimodal reasoning capability of MCAFM is limited, making it difficult to infer high-quality multimodal features for subsequent feature broadcasting, which in turn restricts the overall model performance. When the number of transformer encoders increases to 6, the model achieves a satisfactory performance level. However, further increasing the number to 8 or 10 leads to a substantial rise in computational cost without yielding significant performance gains; in fact, slight declines are observed in some metrics. Therefore, setting the number of transformer encoders in MCAFM to six provides a more reasonable tradeoff between performance and efficiency.

E. Qualitative Results

In this section, we present the segmentation visualization results of MCASEN on the RRSIS-D dataset test set and provide a qualitative analysis. Fig. 7 illustrates MCASEN's segmentation performance across different target types. For large-scale targets such as track fields and golf fields, despite the presence of abundant background regions that are visually similar to the true targets, MCASEN accurately delineates the target areas while maintaining clear boundaries. For medium and small-scale targets such as tennis courts, airplanes, and vehicles, which often appear alongside numerous distractor objects rather than in isolation, the visualizations show that MCASEN can effectively leverage textual information to distinguish between similar objects and accurately segment the true targets indicated by the query text. As shown in the heat map row of Fig. 7, we further visualize the visual features before decoding. The results indicate that the multiscale aggregation and refinement capability of MCRFNet, combined with the role of SPFD in multimodal reasoning, enables the visual features to precisely focus on the target specified by the query text while remaining unaffected by complex background content and cluttered background objects.

To assess MCASEN's ability to handle hard samples, we selected typical difficult cases that frequently caused errors in other methods for visual comparison. As the comparative baseline, we adopted FIANet, which achieved the second-best performance on most metrics, with the results shown in Fig. 8. The hard cases mainly consist of confusing samples and *boundary-blurred samples*. Confusing samples are typ-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

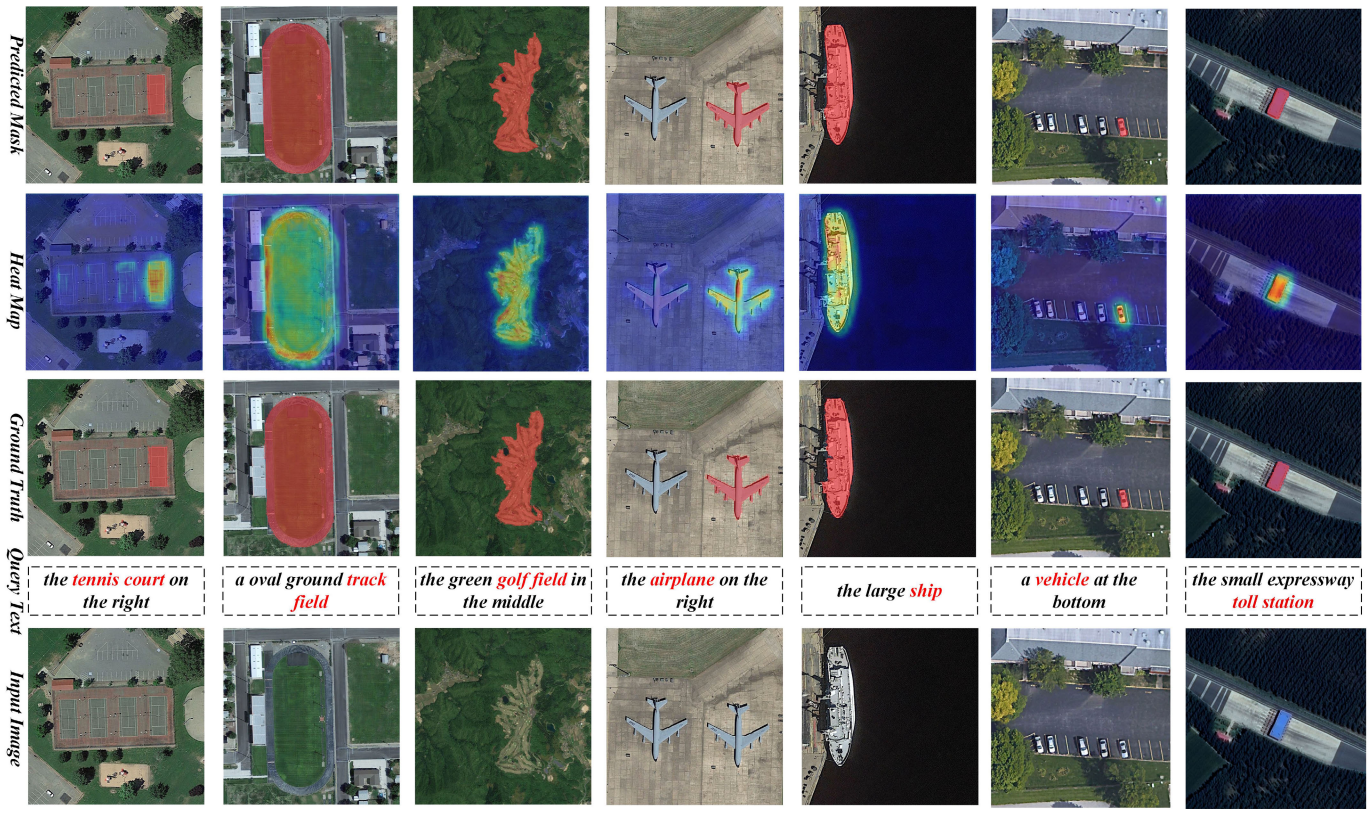


Fig. 7. Visualization results of MCASEN segmentation. (Bottom to top) First row shows the input images, the second row presents the query texts, the third row depicts the ground-truth masks, the fourth row illustrates the feature maps prior to decoding in MCASEN, and the topmost row displays the final segmentation outputs produced by MCASEN. The visualizations demonstrate that MCASEN consistently achieves satisfactory segmentation performance across a variety of target types.

ically small-scale targets surrounded by numerous distractors, whereas boundary-blurred samples tend to blend into the background, increasing segmentation difficulty. As illustrated by the first four samples on the left of Fig. 8, when facing confusing samples, MCASEN benefits from the incorporation of textual context information by SPFD during the decoding stage, enabling it to effectively filter targets matching the text and thereby achieve more precise segmentation. For example, in the second sample from the left, with the query text "a white vehicle on the top," FIANet identifies the vehicle located at the top but fails to capture the attribute "white," mistakenly segmenting the red vehicle instead. In contrast, MCASEN captures key information such as "vehicle," "on the top," and "white," resulting in the correct segmentation of the white vehicle. As shown in the first three samples on the right of Fig. 8, when dealing with boundary-blurred samples, MCRFNet's fine-grained handling of multiscale features effectively preserves target boundaries and detail information, enabling MCASEN to achieve high-quality segmentation even when the target boundaries are indistinct.

Fig. 9 presents the segmentation results of MCASEN across a variety of RS scenes characterized by highly complex backgrounds. As shown, MCASEN successfully identifies and delineates small ground track fields embedded within dense urban structures where small targets are easily confused with surrounding man-made features. In port areas, where linear objects coexist with water-surface reflections and cluttered textures, MCASEN accurately localizes bridges. In desert regions, despite irregular terrain textures and noise-like artifacts, the model effectively detects overpasses. Even in forested environments with substantial vegetation occlusion and high texture similarity, MCASEN is able to clearly segment golf courses. Overall, MCASEN demonstrates robust and precise target localization and segmentation across diverse and challenging background conditions.

We also analyzed bad cases from MCASEN's inference results, as shown in Fig. 10. The results reveal that when RS images contain numerous densely arranged targets, exhibit extremely complex backgrounds, and the query text refers to very small-scale targets, MCASEN struggles to produce accurate segmentation.

In summary, the qualitative results further validate the superiority of the proposed MCASEN: it consistently achieves stable and competitive segmentation performance, whether facing targets of different scales, confusing targets, or boundary-blurred targets.

V. Conclusion

In this article, we address key challenges in the RRSIS task, particularly the lack of semantic guidance during the decoding stage and the coarse design of multiscale feature fusion strategies. We propose the MCASEN, which employs refined multiscale feature fusion and an optimized multimodal reasoning pipeline to effectively mitigate information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

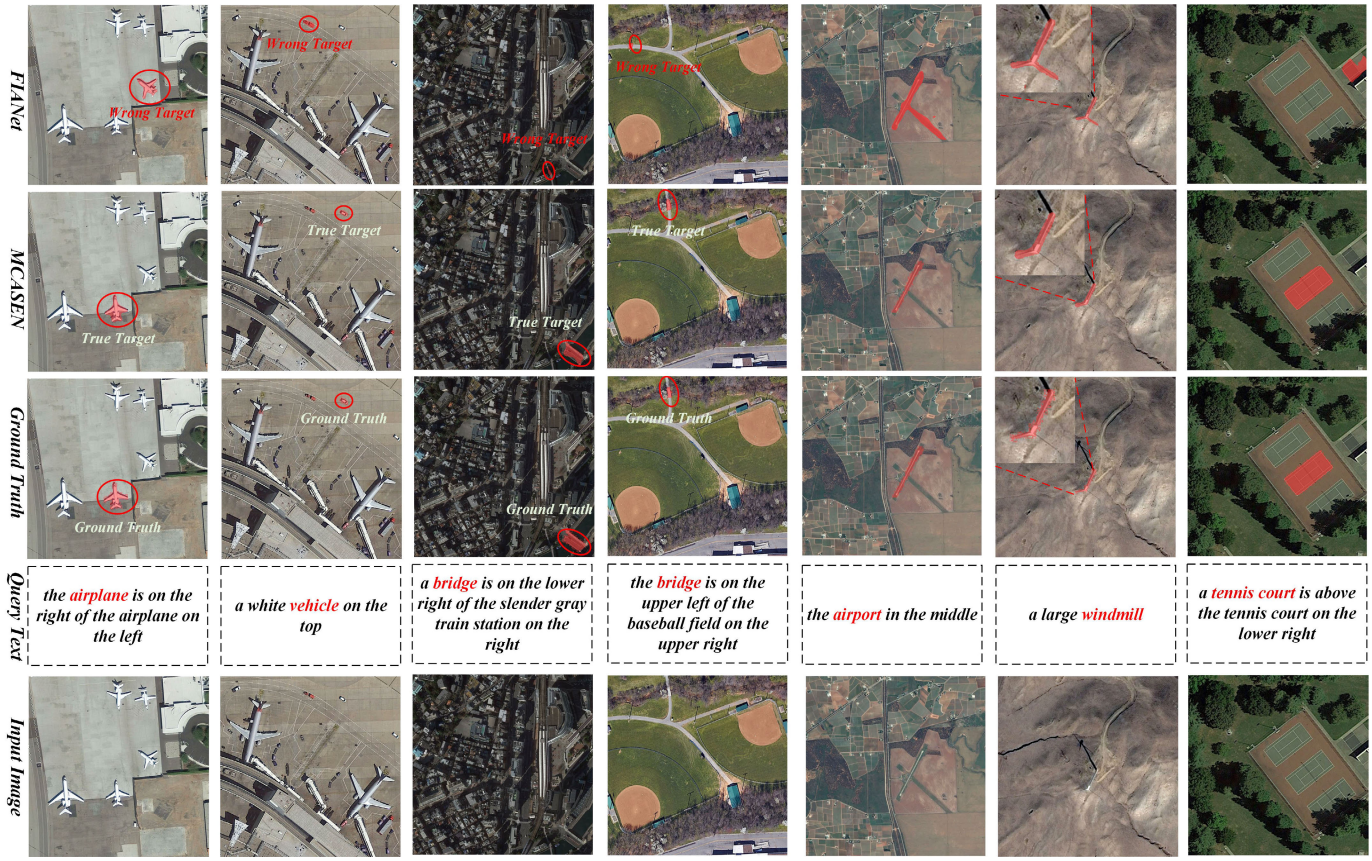


Fig. 8. Segmentation visualization results of MCASEN on hard samples. The figure compares the outputs of MCASEN with those of FIANet, the model that ranked second in most evaluation metrics in the comparative experiments. The samples shown are ones that commonly lead to segmentation errors in the majority of models. (Top to bottom) First row presents the segmentation results of FIANet, and the second row presents those of MCASEN.

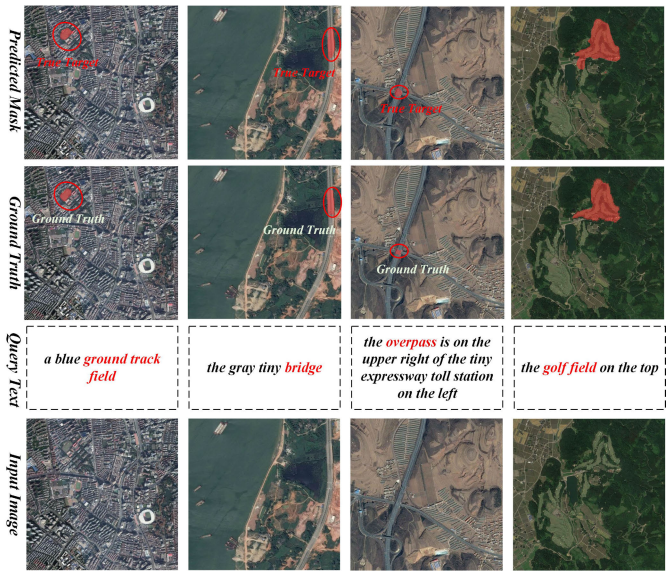


Fig. 9. Segmentation visualization results of MCASEN in complex background scenarios.

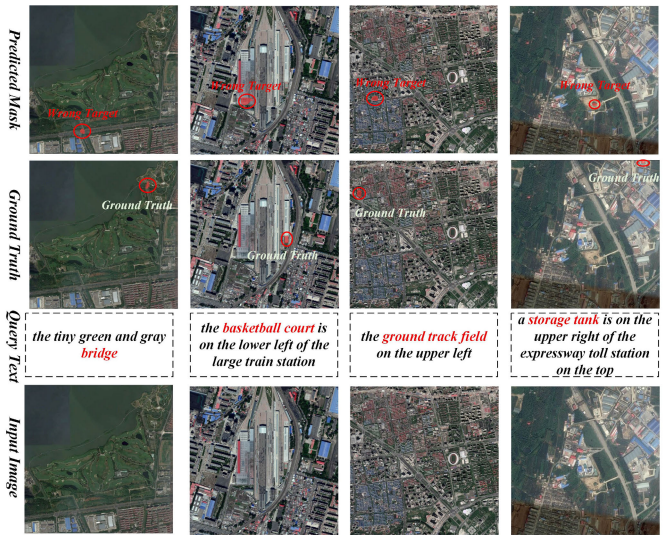


Fig. 10. Bad cases in MCASEN segmentation results. The figure presents visualization examples of erroneous outputs produced by MCASEN. In scenarios where the RS images contain a large number of targets, the background is highly complex, and the query text refers to extremely small-scale targets, MCASEN encounters difficulties in producing accurate segmentation results.

loss and spatial structure degradation, thereby substantially improving segmentation accuracy. MCASEN comprises three core modules: MCRFNet, which utilizes a cross-hierarchical recursive feedback mechanism to refine multiscale feature fusion, significantly reducing the loss of feature information; SPFD, which preserves the spatial structural integrity of visual features during multimodal reasoning and effectively incorpo-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

rates textual contextual cues; and MGFM, which leverages a multipath deep fusion strategy to further enhance the representational quality of visual features and the synergy between multimodal features. Experimental results demonstrate that our method outperforms SOTA approaches across multiple benchmark datasets, thereby validating its effectiveness and superiority. In future work, we plan to further investigate the remaining challenges in RRSIS, including its robustness under highly heterogeneous scenes and its ability to interpret complex or ambiguous textual expressions. We will explore more fine-grained semantic parsing mechanisms and enhance multimodal reasoning to improve adaptability in real-world scenarios. In addition, we aim to integrate RRSIS with large-scale vision–language models to leverage their stronger generalization capability across diverse domains.

REFERENCES

[1] Z. Yuan, L. Mou, Y. Hua, and X. X. Zhu, "RRSIS: Referring remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5613312.

[2] S. Liu et al., "Rotated multi-scale interaction network for referring remote sensing image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26658–26668.

[3] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[4] D. Rolnick et al., "Tackling climate change with machine learning," *ACM Comput. Surv.*, vol. 55, no. 2, pp. 1–96, 2022.

[5] S. N. MohanRajan, A. Loganathan, and P. Manoharan, "Survey on land use/land cover (LU/LC) change analysis in remote sensing and GIS environment: Techniques and challenges," *Environ. Sci. Pollut. Res.*, vol. 27, no. 24, pp. 29900–29926, Aug. 2020.

[6] G. M. Foody, "Remote sensing of tropical forest environments: Towards the monitoring of environmental resources for sustainable development," *Int. J. Remote Sens.*, vol. 24, no. 20, pp. 4035–4046, Jan. 2003.

[7] S. Wang, Y. Han, J. Chen, Z. Zhang, G. Wang, and N. Du, "A deep-learning-based sea search and rescue algorithm by UAV remote sensing," in *Proc. IEEE CSAA Guid., Navigat. Control Conf. (CGNCC)*, Aug. 2018, pp. 1–5.

[8] L. Duan and F. Lafarge, "Towards large-scale city reconstruction from satellites," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2016, pp. 89–104.

[9] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[10] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.

[11] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.

[12] T. Wellmann et al., "Remote sensing in urban planning: Contributions towards ecologically sound policies?," *Landscape Urban Planning*, vol. 204, Dec. 2020, Art. no. 103921.

[13] R. Hu, M. Rohrbach, and T. Darrell, "Segmentation from natural language expressions," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2016, pp. 108–124.

[14] C. Liu, Z. Lin, X. Shen, J. Yang, X. Lu, and A. Yuille, "Recurrent multimodal interaction for referring image segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1271–1280.

[15] Y. Jing, T. Kong, W. Wang, L. Wang, L. Li, and T. Tan, "Locate then segment: A strong pipeline for referring image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9858–9867.

[16] Z. Wang et al., "CRIS: CLIP-driven referring image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11686–11695.

[17] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[18] C. Li et al., "CasFormer: Cascaded transformers for fusion-aware computational hyperspectral imaging," *Inf. Fusion*, vol. 108, Aug. 2024, Art. no. 102408.

[19] Y. Ma et al., "Remote sensing big data computing: Challenges and opportunities," *Future Gener. Comput. Syst.*, vol. 51, pp. 47–60, Oct. 2015.

[20] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.

[21] Z. Zhang, W. Zhang, M. Yan, X. Gao, K. Fu, and X. Sun, "Global visual feature and linguistic state guided attention for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615216.

[22] W. Zhang et al., "Hypersphere-based remote sensing cross-modal text–image retrieval via curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5621815.

[23] M. Li, S. Zang, B. Zhang, S. Li, and C. Wu, "A review of remote sensing image classification techniques: The role of spatio-contextual information," *Eur. J. Remote Sens.*, vol. 47, no. 1, pp. 389–411, Jan. 2014.

[24] H. Ning, B. Zhao, and Y. Yuan, "Semantics-consistent representation learning for remote sensing image–voice retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4700614.

[25] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Expert Syst. Appl.*, vol. 169, May 2021, Art. no. 114417.

[26] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4404119.

[27] S. Lobry, D. Marcos, J. Murray, and D. Tuia, "RSVQA: Visual question answering for remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8555–8566, Dec. 2020.

[28] Z. Yuan, L. Mou, Q. Wang, and X. X. Zhu, "From easy to hard: Learning language-guided curriculum for visual question answering on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623111.

[29] J. Feng, H. Huang, J. Zhang, W. Dong, D. Zhang, and L. Jiao, "SA-MixNet: Structure-aware mixup and invariance learning for scribble-supervised road extraction in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602214.

[30] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[31] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.

[32] C. D. Gilbert and W. Li, "Top-down influences on visual processing," *Nature Rev. Neurosci.*, vol. 14, no. 5, pp. 350–363, May 2013.

[33] R. Li et al., "Referring image segmentation via recurrent refinement networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5745–5753.

[34] V. Nagaraja, V. I. Morariu, and L. S. Davis, "Modeling context between objects for referring expression understanding," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2016, pp. 792–807.

[35] L. Ye, M. Rochan, Z. Liu, and Y. Wang, "Cross-modal self-attention network for referring image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10494–10503.

[36] Z. Hu, G. Feng, J. Sun, L. Zhang, and H. Lu, "Bi-directional relationship inferring network for referring image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4424–4433.

[37] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[38] Z. Yang, J. Wang, Y. Tang, K. Chen, H. Zhao, and P. H. S. Torr, "LAVT: Language-aware vision transformer for referring image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18134–18144.

[39] N. Kim, D. Kim, S. Kwak, C. Lan, and W. Zeng, "ReSTR: Convolution-free referring image segmentation using transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18124–18133.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.

[40] Y. Sun, S. Feng, X. Li, Y. Ye, J. Kang, and X. Huang, "Visual grounding in remote sensing images," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 404–412.

[41] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[42] K. Li, D. Wang, H. Xu, H. Zhong, and C. Wang, "Language-guided progressive attention for visual grounding in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[43] C. Li et al., "Injecting linguistic into visual backbone: Query-aware multimodal fusion network for remote sensing visual grounding," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[44] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, vol. 1. Minneapolis, MI, USA: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186.

[45] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[46] T. Hui et al., "Linguistic structure guided context modeling for referring image segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 59–75.

[47] S. Huang et al., "Referring image segmentation via cross-modal progressive comprehension," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10488–10497.

[48] S. Liu, T. Hui, S. Huang, Y. Wei, B. Li, and G. Li, "Cross-modal progressive comprehension for referring segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 4761–4775, Sep. 2022.

[49] Z. Xu, Z. Chen, Y. Zhang, Y. Song, X. Wan, and G. Li, "Bridging vision and language encoders: Parameter-efficient tuning for referring image segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17457–17466.

[50] Y. Cho, H. Yu, and S.-J. Kang, "Cross-aware early fusion with stage-divided vision and language transformer encoders for referring image segmentation," *IEEE Trans. Multimedia*, vol. 26, pp. 5823–5833, 2024.

[51] Y. Hu et al., "Beyond one-to-one: Rethinking the referring image segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4044–4054.

[52] J. Wu, X. Li, X. Li, H. Ding, Y. Tong, and D. Tao, "Towards robust referring image segmentation," *IEEE Trans. Image Process.*, vol. 33, pp. 1782–1794, 2022.

[53] S. Lei, X. Xiao, T. Zhang, H.-C. Li, Z. Shi, and Q. Zhu, "Exploring fine-grained image-text alignment for referring remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5604611.

[54] M. Liu, X. Jiang, and X. Zhang, "CADFormer: Fine-grained cross-modal alignment and decoding transformer for referring remote sensing image segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 14557–14569, 2025.

[55] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

[56] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, vol. 9, 2010, pp. 249–256.

[57] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[58] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229.

[59] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[60] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

[61] S. Liu, D. Huang, and Y. Wang, "Learning spatial fusion for single-shot object detection," 2019, *arXiv:1911.09516*.

[62] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.



Chongyang Li received the B.Sc. degree from Dalian University of Technology, Dalian, China, in 2022. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include multimodal remote sensing image interpretation and multimodal signal processing.



Wenkai Zhang (Member, IEEE) received the B.Sc. degree from China University of Petroleum, Qingdao, China, in 2013, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2018.

He is currently an Assistant Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include remote sensing image semantic segmentation and multimodal information processing.



Hanbo Bi (Graduate Student Member, IEEE) received the B.Sc. degree from Hunan University, Changsha, China, in 2021. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision, pattern recognition, especially on remote sensing image understanding, semantic segmentation, and few-shot learning.



Jihao Li (Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2017, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2022.

He is currently an Assistant Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image interpretation.



Qiwei Yan received the B.Eng. degree from Beijing Institute of Technology, Beijing, China, in 2022. He is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include computer vision, pattern recognition, and remote sensing image understanding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.



Chen Wang received the B.E. and Ph.D. degrees in electronics and information engineering from Harbin Institute of Technology, Harbin, China, in 2019 and 2025, respectively.



Zhengyi Wang (Member, IEEE) received the B.Sc. degree from China University of Petroleum, Beijing, China, in 2017, the M.Sc. degree from the Ocean University of China, Qingdao, China, in 2020, and the Ph.D. degree from Shanghai Jiao Tong University, Shanghai, China, in 2024.

He is currently an Assistant Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include computer vision and intelligent interpretation of remote sensing images.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.



Hongqi Wang (Member, IEEE) received the B.Sc. degree from Changchun University of Science and Technology, Changchun, China, in 1983, the M.Sc. degree from Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences, Changchun, in 1988, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 1994.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image understanding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:26:45 UTC from IEEE Xplore. Restrictions apply.